

Acadian redfish

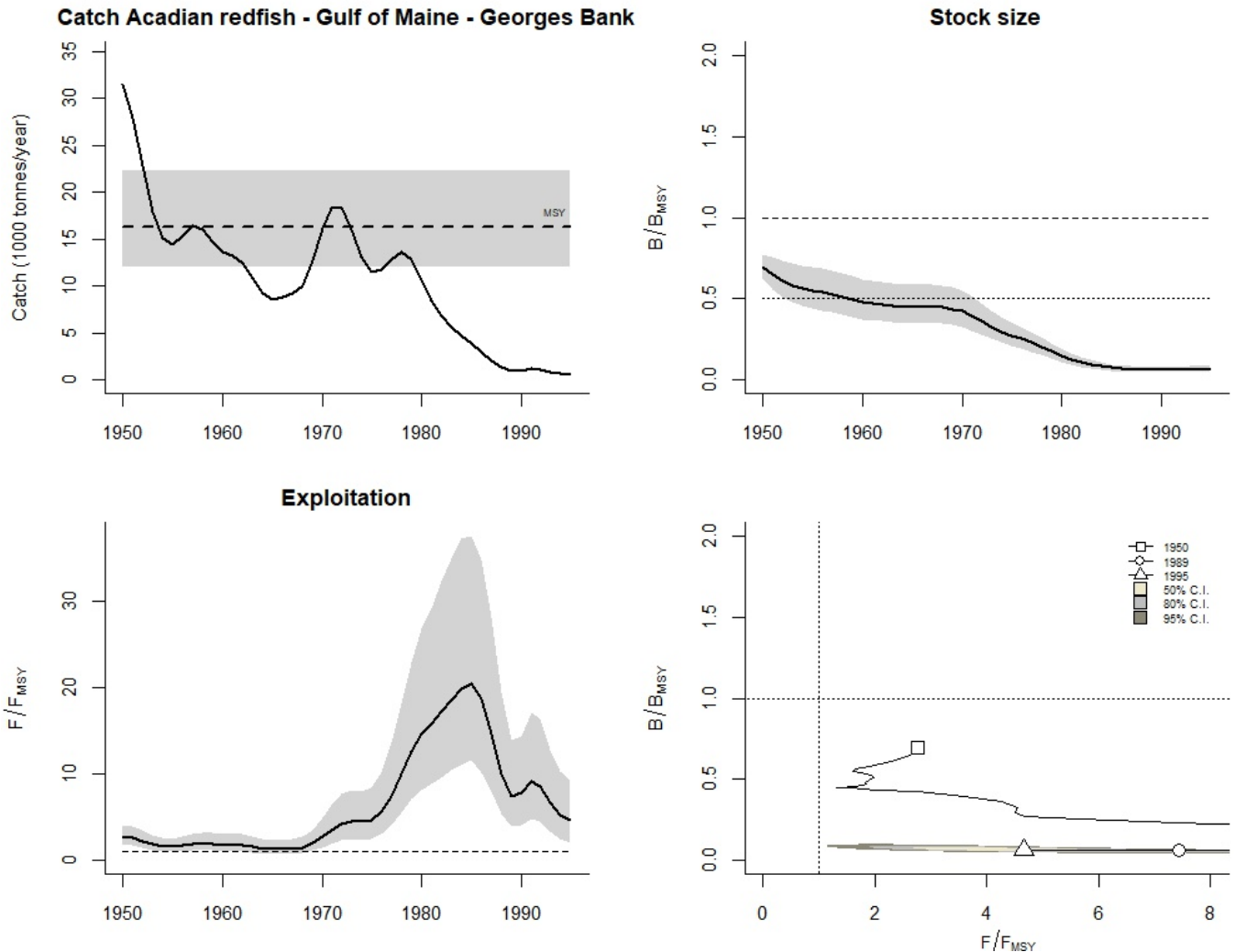
Species: *Sebastes fasciatus*, Stock code: Acadian redfish - Gulf of Maine - Georges Bank

Region: Northeast Shelf

Marine Ecoregion: Gulf of Maine - Georges Bank

Reconstructed catch data used from years 1950 - 1995

For figure captions and method see <http://www.seaaroundus.org/cmsy-method>



extext

Results for management (based on BSM analysis)

$F_{msy} = 0.0362$, 95% CL = 0.0228 - 0.0558 (if $B > 1/2 B_{msy}$ then $F_{msy} = 0.5 r$)

$F_{msy} = 0.0047$, 95% CL = 0.00295 - 0.00723 (r and F_{msy} are linearly reduced if $B < 1/2 B_{msy}$)

$MSY = 16.4$, 95% CL = 12 - 22.2; $B_{msy} = 447$, 95% CL = 316 - 692 (1000 tonnes)

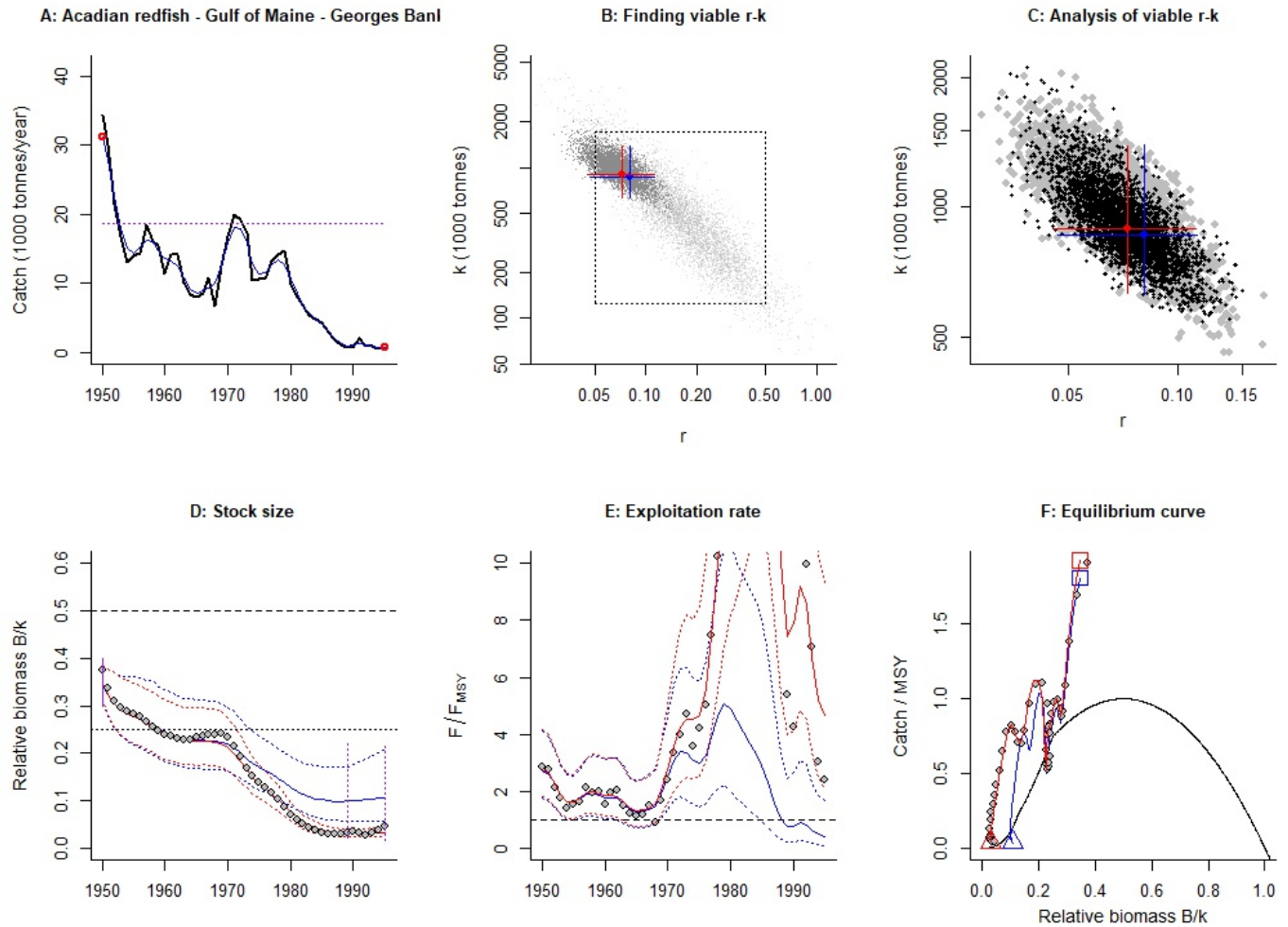
Biomass in last year = 29.2, 95% CL = 20.3 - 45.7 (1000 tonnes)

B/B_{msy} in last year = 0.0648, 95% CL = 0.0475 - 0.0903

Fishing mortality in last year = 0.0219, 95% CL = 0.0128 - 0.0348

$F/F_{msy} = 4.66$, 95% CL = 2.09 - 9.35

Comment: NOAA Assessment model ASAP



extext

Results of CMSY analysis conducted in JAGS

$r = 0.0811$, 95% CL = 0.0467 - 0.113; $k = 859$, 95% CL = 629 - 1397 (1000 tonnes)

MSY = 17.4, 95% CL = 12.5 - 23.4 (1000 tonnes/year)

Relative biomass last year = 0.107 k , 95% CL = 0.0555 - 0.208

Exploitation $F/(r/2)$ in last year = 0.474

Results from Bayesian Schaefer model using catch and CPUE

$r = 0.0724$, 95% CL = 0.0456 - 0.112; $k = 893$, 95% CL = 633 - 1384

r - k log correlation = -0.741

MSY = 16.4, 95% CL = 12 - 22.2 (1000 tonnes/year)

Relative biomass in last year = 0.107 k , 95% CL = 0.0555 - 0.208

Exploitation $F/(r/2)$ in last year = 0.186

$q = 0.0667$, 95% CL = 0.047 - 0.0868

Prior range of q = 0.0117 - 0.645

Relative abundance data type = CPUE

Prior initial relative biomass = 0.3 - 0.4 expert

Prior intermediate relative biomass = 0.0207 - 0.223 in year 1989 default

Prior final relative biomass = 0.0169 - 0.215, default

Prior range for r = 0.05 - 0.5 default, prior range for k = 126 - 1729 (1000 tonnes) default

Source for relative biomass:

NOAA Fisheries. (2020). Assessment Summary Data. Retrieved from www.st.nmfs.noaa.gov/stocksmart. 12/13/2020